

Question number	Answer	Additional guidance	Mark
1(a)	(the membrane is) fluid / fluidity (1)	ACCEPT the phospholipids (and proteins) can move (within the membrane) IGNORE flexible / fluid mosaic model / semipermeable / mosaic / bilayer	(1)

Question number	Answer	Mark
1(b)(i)	The correct answer is A <i>B is incorrect because the units are μm^2</i> <i>C is incorrect because the radius is 150 not 300</i> <i>D is incorrect because the radius is 150 and not 300 and the units are μm^2</i>	(1)

Question number	Answer	Mark
1(b)(ii)	The correct answer is C <i>A is incorrect because the surface area will increase not decrease</i> <i>B is incorrect because the volume will stay the same</i> <i>D is incorrect because surface area will increase not stay the same</i>	(1)

Question number	Answer	Additional guidance	Mark
1(b)(iii)	An explanation that includes two of the following points: - because there is a higher concentration (of oxygen) {outside / in the water} (than in the cytoplasm) (1) - because {all parts of the amoeba are close to the cell membrane / amoeba has a small volume / because it has a large surface area (to volume ratio)} 1) - low energy requirements / low metabolism / low respiratory rate (1)	ACCEPT converse for cytoplasm compared to water ACCEPT partial pressure ACCEPT amoeba is small / diffusion distance is small (to all parts of cell)	(2)